

Autumn Semester examination 2023

Subject: Mathematical Physics II
Full Marks: 40

Subject Code: PHYUGCC05
Time: 2 hrs

[Each question carries 5 marks, Answer any eight questions]

1. Using the definition or by using the generating function of Hermite polynomial $H_n(x)$, prove that

$$(a) H_n(-x) = (-1)^n H_n(x)$$

[2+3]

$$(b) \int_{-\infty}^{+\infty} e^{-x^2} H_n(x) H_m(x) dx = 0; (n \neq m)$$

2. Find the Fourier series corresponding to the periodic function $f(x) = x \sin x$ defined in the interval $-\pi < x < +\pi$ and hence deduce that

$$\frac{\pi}{4} = \frac{1}{2} + \frac{1}{1.3} - \frac{1}{3.5} + \frac{1}{5.7} - \dots$$

3. Assume $+q$ and $-q$ charges are placed at $z = -a$ and $z = +a$ respectively. Prove that the expansion of the potential due to such linear electric dipole can be written in terms of Legendre polynomial. (5)

4. Consider two particles moving along X-axis. In terms of their coordinates x_1 and x_2 , their velocities are given as

$$\frac{dx_1}{dt} = x_2 - x_1 \quad \text{and} \quad \frac{dx_2}{dt} = x_1 - x_2$$

respectively. When they start moving from their initial conditions $x_1(0) = 1$ and $x_2(0) = -1$, the time dependence of both x_1 and x_2 contain a term of the form e^{at} . Find a . (5)

5. Prove that the Legendre polynomials $P_n(x)$ can directly be generated from the Rodrigue's formula as, (5)

$$P_n(x) = \frac{1}{2^n n!} \frac{d^n}{dx^n} (x^2 - 1)^n$$

6. Consider the Laplace differential equation for a three-dimensional cylindrical polar coordinate system (r, θ, φ) as,

$$\frac{\partial^2 u}{\partial r^2} + \frac{1}{r} \frac{\partial u}{\partial r} + \frac{1}{r^2} \frac{\partial^2 u}{\partial \varphi^2} + \frac{\partial^2 u}{\partial z^2} = 0$$

Adopting the separation of variable technique, find the general solution $u(r, \varphi, z)$ of the differential equation. (5)

7. Consider Bessel's differential equation of the form $x^2 y'' + xy' + (x^2 - n^2)y = 0$, Test the singularity. Using the Frobenius-Fuch theorem, is it possible to get two linearly independent solutions of the above-mentioned equation? Justify your answer. (1+2+2)

8. A function $f(x)$ is expanded in a series $f(x) = \sum_{n=0}^{\infty} a_n P_n(x)$; where $P_n(x)$ defines the Legendre polynomial of order n . Show that (5)

$$\int_{-1}^{+1} [f(x)]^2 dx = \sum_{n=0}^{\infty} \frac{2a_n^2}{2n+1}$$

(2.5+2.5)

9. Prove the following identities:

$$(a) \beta(m, n) = \frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)}$$

$$(b) \Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$$

where the beta function $\beta(m, n)$ and gamma function $\Gamma(n)$ respectively defined as,

$$\beta(m, n) = \int_0^1 x^{m-1} (1-x)^{n-1} dx \quad \text{and} \quad \Gamma(n) = \int_0^{\infty} e^{-x} x^{n-1} dx$$

10. Consider the Hypergeometric differential equation

(2+3)

$$x(x-1)y'' + [(1+\alpha+\beta)x - \gamma]y' - \alpha\beta y = 0$$

given that α , β and γ all are constants and γ cannot take negative value.

- (a) Examine the singularity of the above differential equation at $x = 0$.
(b) Using the ascending power series of the form $y = \sum_{k=0}^{\infty} a_k x^{k+\lambda}$, determine the two roots of the differential equation i.e. two possible values of λ (not the full solution).
11. Establish the relation:

$$J_n(x)J'_{-n}(x) - J_{-n}(x)J'_n(x) = -\frac{2 \sin n\pi}{\pi x}$$

$J_n(x)$ represents Bessel polynomial of order n .

(5)

Aliah University
Autumn (odd) Semester Examination -2023

Course Code: PHYUGCC06

Course Title: Thermal Physics (Theory)

Department of Appearing Students: PHYSICS, SEM-III, 2nd Year

Full Marks: 40

Credit: 04

Time: 2.00 hrs

[Use separate sheet for answering Group-A and Group-B questions.]

Group-A [Attempt any FOUR questions]

1. The velocity distribution function of a group of N particles is given by

$$dN_v = kvdv \quad \text{for } 0 \leq v \leq V$$

$$= 0 \quad \text{for } v > V$$

Calculate the constant k in terms of N and V and also calculate the mean velocity and RMS velocity of the particle. [2+1.5+1.5]

2. Maxwell's law of velocity distribution for a 2-dimensional ideal gas is given by

$$F(c)dc = \frac{m}{kT} e^{-\frac{mc^2}{2kT}} cdc,$$

notations have their usual meaning. Indicate graphically how the distribution changes with change in temperature. From the above distribution law, find the expression for (i) mean velocity, (ii) most probable velocity. [1+2+2]

3. (a) For a gas composed of diatomic molecules with translational and rotational degrees of freedom, calculate the ratio of specific heats (γ) using the equipartition principle. Express the result in terms of the ideal gas constant (R).

(b) In 1 - dimension, an ensemble of N classical particles has energy of the form, $E = \frac{1}{2m}(p_x^2) + \frac{1}{2}kx^2$, where m is the mass of the particle, k is a constant. Find the average internal energy of the system at temperature T . [2.5+2.5]

4. (a) An ideal gas expands adiabatically so that its volume is doubled. How many times will the number of collisions per second of the molecules decrease? (given, $\gamma=1.4$).

(b) A narrow molecular beam makes its way into a vessel filled with gas under low pressure. Find the mean free path of the molecules if the beam intensity decreased to $1/n$ times its initial intensity over a distance Δl . [2.5+2.5]

5. (a) Write down the van der Waals equation of state and show that with the increase of volume per mole, the van der Waals equation of state tends to that of an ideal gas.

(b) What is Boyle temperature? How is it related to critical temperature? Calculate the ratio of Boyle temperature to critical temperature? [(1+1)+3]

6. (a) 'Brownian motion may be considered as analogous to the thermal motion of the molecules but on much reduced scale' - explain the significance of the statement.

(b) Write down the expression (no derivation) for co-efficient of viscosity and explain, how does the co-efficient of viscosity (η) vary with Pressure and temperature? [2.5+2.5]

Group-B [Attempt any FOUR questions]

7. Define- Enthalpy (H), Helmholtz free energy (F), Gibbs free energy (G), entropy (S). Hence, derive the following Maxwell's thermodynamic relations:

$$(a) \left(\frac{\partial T}{\partial V}\right)_S = -\left(\frac{\partial P}{\partial S}\right)_V$$

$$(b) -\left(\frac{\partial S}{\partial P}\right)_T = \left(\frac{\partial V}{\partial T}\right)_P$$

[2+3=5]

8. (a) State Kelvin-Planck statement of the second law of thermodynamics.

(b) With appropriate P-V indicator diagram obtain an expression for the efficiency of a reversible Carnot engine. [2+3=5]

9. (a) State and prove Carnot's theorem.

(b) An inventor from Aliah University claims to have developed an engine working between 650 K and 330 K capable of having an efficiency of 55%. Comment on his claim. [3+2=5]

10. (a) Derive the Clausius-Clapeyron equation: $(dp/dT)_{\text{sat}} = L/T(V_{\text{vap}} - V_{\text{liq}})$.

(b) Assuming the specific heat capacity of water as $S=1 \text{ k cal/kg/}^\circ\text{C}$, find the change in entropy when 0.15 kg of water at 35°C is mixed with .55 kg of water at 55°C . [3+2=5]

11. Define extensive and intensive variable. Make tabular list of extensive and intensive variables for the given quantities: volume (V), length (L), area (A), charge (Q), magnetization (I), pressure (P), tension (F), surface tension (S), EMF (E), magnetic intensity (H), density (D). [2+3=5]

Odd (Autumn) Semester Examination - 2023
Department of Physics

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Course Code: PHYUGCC07

Course Title: Analog Systems and Applications

Department of Appearing Students: PHYSICS, SEM-III, 2ND Year

Full Marks: 40

Time: 2.00 Hour

Answer any FOUR questions

1. (a) Draw and explain working principle of a Bridge rectifier.
~~(b)~~ Mention the advantages and disadvantages of a bridge rectifier over full wave rectifier.
(c) What is meant by Avalanche breakdown?
5+3+2=10

2. (a) Show the relation between α and β considering transistor leakage currents.
~~(b)~~ Find out the expression for the stability factor for collector to base bias circuit of a transistor in C-E mode.
~~(c)~~ Show that introduction of negative feedback gives better stability in gain for a feedback amplifier.
3+4+3=10

3. ~~(a)~~ Draw the circuit diagram to obtain the input/output characteristic for a transistor operating in C-E mode. Explain the transistor output characteristic in C-E mode.
(b) Find the operating point of a n-p-n transistor working in the C-E configuration with the following specification $V_{CC} = 20\text{ V}$, $R_B = 100\text{ K}\Omega$, $R_C = 1\text{ K}\Omega$, $\beta = 100$ and $V_{BE} = 0.2\text{ V}$.
(3+3)+4=10

4. ~~(a)~~ Find out the expression for the power gain and input resistance of a transistor amplifier operating in the C-E mode using hybrid parameters.
(b) Derive an expression for the frequency of oscillations of a Wien bridge oscillator with circuit diagram
5+5=10

5. ~~(a)~~ What are the characteristic of an ideal OPAMP?
(b) Derive an expression for closed loop gain of an OPAMP as non-inverting amplifier.
(c) Show that OPAMP can be used as differential amplifier.
3+4+3=10

1) centre - tapped transformer not needed
for same DC output,
2) output current smaller of B_{DC} than full wave rectifier.

$$\frac{C_1}{C_1 + C_2}$$

$$A_{D''} = \frac{A_{D'}}{1 + A_{D'}(h_{fe}) \times R_e}$$

Odd (Autumn) Semester Examination -2023

Course Code: PHYUGSE01

Course Title: Basic Instrumentation Skill

Department of Appearing Students: PHYSICS, SEM-III, 2nd Year

Time: 2.00 hrs

Full Marks: 40

Answer Any Four Question

4 × 10 = 40

1. With schematic diagram of electrostatic deflection system, find the deflection factor of a CRT. Describe the XY mode of a CRO.

7+3

2. Write the construction of permanent magnet moving coil (PMMC) ammeter and describe its torque equation.

10

3. Draw and discuss the Two-terminal method, five-terminal method, and four-terminal-pair method of LCR meter measurement circuit.

10

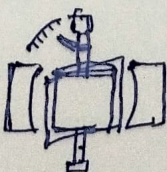
4. Detail the categorization, types, and ranges of vacuum. Briefly explain the operational principles of Rotary and Ion pump.

10

5. Deduce the balance condition for DC and AC bridge with necessary circuit diagram. Examine any one procedure for measuring self-inductance..

5+5

① When the current flow is zero the AC Bridge is in balanced condition



② When the current flow is stop the DC Bridge is in balanced condition

$$T = K B \theta$$

Affusion pump
Mechanical pump

ALIAH UNIVERSITY
END SEMESTER EXAMINATION, AUTUMN 2024
PAPER: MATUGGE03 (Differential Equation and Integral Transformation)

Full Marks: 60

Time Allowed: 2 Hours 30 Minutes

All symbols have their usual meanings
Use different answer script for different group

GROUP - A

Answer any **THREE** of the following questions.

$10 \times 3 = 30$

1. (a) What is exact differential equation? 2

(b) Examine the exactness of the following differential equation and solve them:

i. $(x^2 - y)dx + (y^2 - x)dy = 0.$ 4

ii. $(y^2 e^{xy^2} + 4x^3)dx + (2xye^{xy^2} - 3y^2)dy = 0.$ 4

2. (a) Form a differential equation by eliminating the parameter(λ) from the following relation 5

$$\frac{x^2}{a^2 + \lambda} + \frac{y^2}{b^2 + \lambda} = 1.$$

- (b) Find the general solution and singular solution(if exists) of the differential equation 5

$$y = px + \sqrt{a^2 p^2 + b^2}, \quad p = \frac{dy}{dx}.$$

3. (a) Solve the homogeneous differential equation

$$\frac{dy}{dx} = \frac{y}{x} + \tan \frac{y}{x}$$

- (b) If

$$\frac{dr}{d\theta} + 2r \tan \theta = \sin \theta$$

and $r = 0$, when $\theta = \frac{\pi}{3}$, show that the maximum value of r is $\frac{1}{8}$.

4. (a) Solve the differential equation 6

$$x^2 \frac{d^2 y}{dx^2} - 3x \frac{dy}{dx} + 4y = 2x^2.$$

- (b) Find particular integral of 4

$$\frac{d^2 y}{dx^2} - 2 \frac{dy}{dx} = e^x \sin x.$$

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$$\frac{d^2 y}{dx^2} - 2 \frac{dy}{dx} = e^x \sin x.$$

5. (a) Solve by the method of variation of parameter

6

$$\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + y = \frac{e^{-x}}{x^2}.$$

- (b) Find particular integral of

4

$$(D^3 - 2D^2 - 5D + 6)y = (e^{2x} + 3)^2.$$

GROUP - B

Answer any **THREE** questions from question no 1 to 5.

5 × 3 = 15

1. By eliminating three arbitrary constants a, b, c from the family of ellipsoids

5

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1,$$

Obtain

$$zx \frac{\partial^2 z}{\partial x^2} + x \left(\frac{\partial z}{\partial x} \right)^2 - z \frac{\partial z}{\partial x} = 0,$$

$$zy \frac{\partial^2 z}{\partial y^2} + y \left(\frac{\partial z}{\partial y} \right)^2 - z \frac{\partial z}{\partial y} = 0,$$

$$z \frac{\partial^2 z}{\partial x \partial y} + \frac{\partial z}{\partial x} \frac{\partial z}{\partial y} = 0.$$

2. Eliminate the arbitrary function ϕ from the relation

5

$$\phi \left(\frac{x-a}{z-c}, \frac{y-b}{z-c} \right) = 0.$$

3. Solve the following partial differential equation

5

$$z(x+y)p + z(x-y)q = x^2 + y^2.$$

4. Solve the partial differential equation by using Lagrange's Method

5

$$y^2 p - xyq = x(z - 2y).$$

5. Find a complete integral of

5

$$2z + p^2 + qy + 2y^2 = 0.$$

Answer any **THREE** questions from question no 6 to 10.

$5 \times 3 = 15$

6. Define the Laplace Transformation for a function $f(x)$. Can we apply the Laplace Transformation for every function $f(x)$? Justify your answer. By using definition find the Laplace Transformation of $f(x) = x$. 5

7. Using Laplace transformation solve the differential equation

$$y'' + 3y' + 2y = 12e^{2x},$$

with $y(0) = 1$ and $y'(0) = -1$. 5

8. State convolution theorem. Using Convolution theorem evaluate 5

$$\mathcal{L}^{-1} \left\{ \frac{s^2}{(s^2 + 4)^2} \right\}.$$

9. Find the Fourier Cosine and Sine transformation of e^{-bx} and hence evaluate the integral 5

$$\int_0^{\infty} \frac{\alpha \sin \alpha x}{\alpha^2 + b^2} d\alpha.$$

10. Using Convolution theorem for Fourier transformation solve the integral equation

$$\int_{-\infty}^{+\infty} f(y)f(x-y)dy = \frac{4}{x^2 + 4},$$

for all $x \in (-\infty, +\infty)$. 5